

Day 54 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Science & Technology (1)

READING MATERIAL

Covers ISRO's most significant recent achievements: the SpaDeX docking mission (India's 4th-in-the-world space docking capability) and the Gaganyaan human spaceflight programme's progress toward its 2026 target.

Q: What is the SpaDeX mission, when was it launched, and what rocket was used?

A: SpaDeX (Space Docking Experiment) launched on 30 December 2024, using the PSLV-C60 rocket, deploying two small satellites (SDX01 and SDX02) into orbit to demonstrate autonomous docking technology.

Q: When did SpaDeX achieve its first successful docking, and what milestone did this mark for India?

A: 16 January 2025 — making India the 4th country in the world (after the USA, Russia, and China) to successfully demonstrate space docking technology.

Q: When was SpaDeX's SECOND docking demonstration achieved?

A: 20 April 2025.

Q: Why is SpaDeX's docking technology considered a precursor for Chandrayaan-4?

A: Because Chandrayaan-4's payload will be too heavy to launch in a single lift-off — it will require multiple launches and IN-ORBIT DOCKING (the exact capability SpaDeX demonstrated) to assemble the full mission.

Q: What is Gaganyaan, and what is its current target year (revised from an earlier date)?

A: Gaganyaan is ISRO's human spaceflight programme, aiming to send Indian astronauts into orbit. Originally planned for 2024, its target has been pushed to 2026.

Q: What was the IADT-01 test (24 August 2025), and what did it involve?

A: ISRO's first Integrated Air Drop Test for Gaganyaan — a dummy crew capsule weighing about 4.8 tonnes was dropped from roughly 3 km altitude by an IAF Chinook helicopter, descending safely via a sequenced parachute deployment before recovery.

MOCK TEST (WITH ANSWERS)

1. The SpaDeX mission was launched using which rocket?

- A. GSLV Mk III
- B. PSLV-C60 ✓**
- C. SSLV-D2
- D. LVM3

Explanation: PSLV-C60. [medium, web-research]

2. SpaDeX achieved its first successful docking on:

- A. 30 December 2024
- B. 16 January 2025 ✓**
- C. 20 April 2025
- D. 24 August 2025

Explanation: 16 January 2025. [medium, web-research]

3. With SpaDeX's successful docking, India became the ____ country in the world to achieve space docking technology.

- A. 2nd
- B. 3rd
- C. 4th ✓**
- D. 5th

Explanation: 4th (after USA, Russia, China). [medium, web-research]

4. SpaDeX's docking technology is considered a precursor for which future ISRO mission?

- A. Aditya-L1
- B. Mangalyaan-2

C. Chandrayaan-4 ✓

D. NISAR

Explanation: Chandrayaan-4. [medium, web-research]

5. Gaganyaan, ISRO's human spaceflight programme, was originally planned for 2024 but is now targeted for:

A. 2025

B. 2026 ✓

C. 2028

D. 2030

Explanation: 2026. [medium, web-research]

6. What was tested in ISRO's IADT-01 test on 24 August 2025?

A. A satellite launch

B. A dummy crew capsule's parachute-assisted descent ✓

C. A new rocket engine

D. A lunar rover

Explanation: A dummy crew capsule's parachute-assisted descent. [hard, web-research]

7. Approximately how much did the dummy crew capsule weigh in the IADT-01 test?

A. 1.5 tonnes

B. 2.8 tonnes

C. 4.8 tonnes ✓

D. 6.2 tonnes

Explanation: 4.8 tonnes. [hard, web-research]

Part B – Aptitude & Mental Ability: Simple Interest

READING MATERIAL

Simple Interest is 1 of 10 named Aptitude sub-skills. Real exam evidence shows TNPSC favors word-problem variants (doubling time, comparing two loans, fractional rates with month-based time) over the bare 'calculate SI' formula alone.

Q: Formula: What is the Simple Interest formula, and how do you get the doubling time from it?

A: $SI = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$. A sum **DOUBLES** when SI equals the Principal itself: $P = (P \times R \times T) / 100 \rightarrow T = 100/R$ years. So at 8% per annum, doubling time = $100/8 = 12.5$ years = 150 months.

Q: Real exam (2024): In how many months will Rs.2,000 double itself at 8% per annum simple interest?

A: $T = 100/R = 100/8 = 12.5$ years = $12.5 \times 12 = 150$ months.

Q: Real exam (2022): What is the simple interest on Rs.1,000 for one year at 10% per annum?

A: $SI = (1000 \times 10 \times 1) / 100 = \text{Rs.}100$.

Q: Real exam (2024): Find the simple interest on Rs.68,000 at $16\frac{2}{3}\%$ per annum for 9 months.

A: Convert the mixed-fraction rate: $16\frac{2}{3}\% = 50/3\%$. Convert months to years: 9 months = $9/12 = 3/4$ year. $SI = 68,000 \times (50/3) \times (3/4) / 100 = 68,000 \times 12.5/100 = \text{Rs.}8,500$. (The 3s cancel neatly between the rate's denominator and the time's numerator — recognizing that shortcut avoids messy fraction arithmetic.)

Q: Real exam (2025): A borrows Rs.16,000 at 12% p.a. simple interest, B borrows Rs.12,000 at 18% p.a. simple interest. In how many years will their total amounts owed (principal+interest) become equal?

A: Set up the equality: $16000 + 16000(0.12)t = 12000 + 12000(0.18)t \rightarrow 16000 + 1920t = 12000 + 2160t \rightarrow 4000 = 240t \rightarrow t = 16\frac{2}{3}$ years.

Q: Method — finding the rate when given principal, time, and final amount: A sum of Rs.5,000 becomes Rs.6,200 in 3 years at simple interest. Find the rate.

A: $SI = \text{Amount} - \text{Principal} = 6200 - 5000 = 1200$. Rearranged formula: $R = (SI \times 100) / (P \times T) = (1200 \times 100) / (5000 \times 3) = 120000/15000 = 8\%$ per annum.

MOCK TEST (WITH ANSWERS)

1. In how many years will a sum double itself at 5% per annum simple interest?

- A. 10 years
- B. 15 years
- C. 20 years ✓**
- D. 25 years

Explanation: $T = 100/R = 100/5 = 20$ years. [easy, authored]

2. Find the simple interest on Rs.2,500 for 1 year at 12% per annum.

- A. Rs.250
- B. Rs.275
- C. Rs.300 ✓**
- D. Rs.320

Explanation: $SI = (2500 \times 12 \times 1) / 100 = \text{Rs.}300$. [easy, authored]

3. Find the simple interest on Rs.54,000 at $13\frac{1}{3}\%$ per annum for 6 months.

- A. Rs.3,600 ✓**
- B. Rs.3,000
- C. Rs.3,300
- D. Rs.4,000

Explanation: $13\frac{1}{3}\% = 40/3\%$. 6 months = $1/2$ year. $SI = 54000 \times (40/3) \times (1/2) / 100 = 54000 \times 40/600 = 54000 \times (1/15) = 3,600$. [hard, authored]

4. C borrows Rs.20,000 at 10% p.a. simple interest, D borrows Rs.15,000 at 15% p.a. simple interest. In how many years will their total amounts owed become equal?

- A. 8 years
- B. 10 years

C. 20 years ✓

D. 25 years

Explanation: $20000 + 2000t = 15000 + 2250t \rightarrow 5000 = 250t \rightarrow t = 20$ years. [hard, authored]

5. A sum of Rs.8,000 becomes Rs.9,440 in 3 years at simple interest. Find the rate per annum.

A. 5%

B. 6% ✓

C. 6.5%

D. 7%

Explanation: $SI = 9440 - 8000 = 1440$. $R = (1440 \times 100) / (8000 \times 3) = 144000 / 24000 = 6\%$. [medium, authored]

6. What sum will yield Rs.900 as simple interest in 3 years at 6% per annum?

A. Rs.4,000

B. Rs.5,000 ✓

C. Rs.5,500

D. Rs.6,000

Explanation: $P = (SI \times 100) / (R \times T) = (900 \times 100) / (6 \times 3) = 90000 / 18 = \text{Rs.}5,000$. [medium, authored]

Part C – General English: Prose: "The Tempest" – Tales From Shakespeare

READING MATERIAL

Charles and Mary Lamb's prose retelling of Shakespeare's play: Prospero, the overthrown Duke of Milan, and his daughter Miranda live marooned on an island with the spirit Ariel and the enslaved Caliban — until Prospero conjures a storm to bring his old enemies to the island, setting up reconciliation rather than revenge.

Q: Real exam: Is it true that 'The Tempest' was written in 1500? What IS true about when it was written, relative to Shakespeare's career?

A: NO — 'written in 1500' is FALSE (Shakespeare actually wrote it around 1610-1611). What IS true, confirmed by a real exam: it is considered one of the LAST plays William Shakespeare wrote.

Q: Real exam: Is 'The Tempest' believed to be based on a real event?

A: Yes — confirmed true by a real exam question: it is believed to be based on an ACTUAL shipwreck (commonly identified as the 1609 wreck of the Sea Venture off Bermuda).

Q: Where do Prospero and Miranda live, and why did they end up there?

A: In a cave on an isolated island. They ended up there because Prospero, the rightful Duke of Milan, was EXILED by his brother Antonio, who usurped his dukedom.

Q: Who is Ariel, who is Caliban, and what does Prospero use to control the spirits and elements?

A: Ariel is a spirit FREED by Prospero (from imprisonment by the witch Sycorax) and made obedient to him. Caliban is enslaved by Prospero. Prospero controls the spirits and elements using a magic WAND.

Q: Why did Prospero raise the storm (the 'tempest') at sea?

A: To bring his enemies — the King of Naples and his own brother Antonio — to the island, setting up the story's resolution.

Q: Real exam: Who is Prospero's daughter, and what game did she play with Ferdinand on the island?

A: Miranda is Prospero's daughter — confirmed by a real exam question. She and Ferdinand played CHESS together on the island — also confirmed by a real exam question.

Q: What task did Prospero assign to Ferdinand, why, and what was the relationship between Ferdinand and Miranda?

A: Prospero assigned Ferdinand the hard labour of PILING LOGS OF WOOD — a deliberate test of Ferdinand's love and devotion to Miranda. Miranda and Ferdinand admired each other (the seeds of a romantic relationship).

Q: How did Ariel torment the King of Naples and Antonio, and how did they feel afterward?

A: By creating illusions and strange sounds. Afterward, they felt sorry and repentant — setting up Prospero's eventual forgiveness rather than revenge.

Q: What did Prospero do before finally leaving the island?

A: He forgave everyone (his former enemies) and set Ariel free.

Q: Who helped Prospero survive when he was first cast out to sea with infant Miranda?

A: A kind lord named Gonzalo.

MOCK TEST (WITH ANSWERS)

1. Which of the following is TRUE about the play 'The Tempest'?

A. It was written in 1500

B. It is considered one of Shakespeare's last plays ✓

C. It was Shakespeare's very first play

D. It has no connection to any real historical event

Explanation: One of Shakespeare's last plays. [medium, question-bank-2025]

2. 'The Tempest' is believed to be based on:

A. A real royal wedding

B. An actual shipwreck ✓

C. A real war between Naples and Milan

D. A real volcanic eruption

Explanation: An actual shipwreck. [medium, question-bank-2025]

3. Where did Prospero and Miranda live?

- A. In a palace
- B. In a cottage in the woods
- C. In a cave on an island ✓**
- D. In a ship

Explanation: In a cave on an island. [easy, web-research]

4. Who was Ariel in the story?

- A. A monster enslaved by Prospero
- B. The King of Naples
- C. A spirit freed by Prospero ✓**
- D. Prospero's brother

Explanation: A spirit freed by Prospero. [medium, web-research]

5. Why did Prospero and Miranda end up on the island?

- A. They were shipwrecked by accident
- B. They were exiled by Antonio ✓**
- C. They fled from a war
- D. They were exploring the sea

Explanation: They were exiled by Antonio. [medium, web-research]

6. What did Prospero use to control the spirits and the elements?

- A. A magic ring
- B. A spell book
- C. A magic wand ✓**
- D. A special potion

Explanation: A magic wand. [medium, web-research]

7. What task did Prospero assign to Ferdinand as a test of his love for Miranda?

- A. Guarding the cave
- B. Piling logs of wood ✓**
- C. Finding food for Miranda
- D. Building a ship

Explanation: Piling logs of wood. [medium, web-research]

8. What game did Ferdinand and Miranda play together on the island?

- A. Carrom
- B. Hide and Seek
- C. Chess ✓**
- D. Cards

Explanation: Chess. [hard, question-bank-2024]

9. How did Ariel torment the King of Naples and Antonio?

- A. By sending them back to sea
- B. By pretending to be a monster
- C. By imprisoning them in a cave
- D. By creating illusions and strange sounds ✓**

Explanation: By creating illusions and strange sounds. [medium, web-research]

10. What did Prospero do before finally leaving the island?

- A. Destroyed the island
- B. Punished his enemies
- C. Forgave everyone and set Ariel free ✓**
- D. Took over the island as king

Explanation: Forgave everyone and set Ariel free. [medium, web-research]